Efficient direct localization of overlapping wideband sources with unequal powers: multi-candidate joint refinement and waveform-eliminated updates

ARTICLE INFO

Keywords:
Direct position determination
Overlapping sources
Alternating projection
Block NOMP
Variable projection

ABSTRACT

Localizing unknown wideband emitters from mixed observations becomes difficult when their spectra overlap and their received powers differ. Direct position determination fits the station observations jointly, but its reliability and cost depend on candidate selection and position refinement. This paper studies two improvements for synchronized single-antenna stations under a known relative-amplitude and baseband-delay model. Multi-candidate joint-refinement alternating projection (MCJ-AP) retains several conditional candidates, jointly refines both positions for each branch, and selects the smallest final residual. Waveform-eliminated block Newtonized orthogonal matching pursuit (VP-BNOMP) re-estimates the current source coefficients at each trial position and uses the resulting reduced curvature. Across 2,400 observations from 20 source configurations, five-candidate MCJ-AP increases the two-source success rate within 20 m from 95% to 99%, with the recoveries concentrated in one configuration. At matched candidates and stopping accuracy, joint refinement reduces mean complete runtime by 31.36%. In 240 paired observations, reduced curvature lowers mean runtime by 83.77% relative to fixed-coefficient curvature with the same waveform-eliminated acceptance objective. On 480 observations from 16 new source configurations, both single- and five-candidate AP succeed throughout, whereas VP-BNOMP succeeds in 81.25%. The results identify the conditional reliability benefit of candidate retention and the computational benefit of removing position–waveform coupling from local updates.

1. Introduction

Emitter localization often estimates time differences of arrival before solving for position. Generalized correlation and hyperbolic localization provide established tools for this two-step approach [3, 8]. When unknown emitters occupy the same frequency band, each station receives a mixture with different propagation responses. Extracting and associating individual delays then becomes difficult, particularly for a weak emitter in the presence of a stronger one. Direct position determination (DPD) incorporates propagation into the observation model and estimates positions from the station data jointly [5, 13].

This paper considers two independent unknown wideband sources observed by four synchronized single-antenna stations. The positions, transmitted waveforms and powers are unknown;

station coordinates and relative propagation responses are available. A single-source spectrum can assign several large peaks to the stronger source, whereas an exhaustive search over two two-dimensional positions requires many combinations. Sequential discovery followed by continuous refinement reduces this search, but leaves two decisions: which candidate regions to retain, and how to update positions while estimating the unknown waveforms.

Wideband maximum likelihood and alternating projection (AP) reduce joint search dimensionality by updating one position conditional on the others [5]. MUSIC-type DPD uses orthogonality to an estimated noise subspace [13]. MVDR uses the inverse covariance to suppress interfering components [2], and DPD-HR applies this principle to direct localization [12]. These methods process mixtures differently, and their relative performance also depends on covariance estimation and search. We therefore compare concrete implementations using the same data and propagation information.

Newtonized orthogonal matching pursuit (NOMP) combines greedy discovery, Newton refinement and cyclic updates [10]. The two-dimensional multi-frequency block NOMP of Yang et al. groups the unknown frequency coefficients at a common position and re-estimates them after position updates [14]. It supplies a close reference for the local update studied here. QNOMP eliminates path gains and jointly refines delays and angles from pilot-based channel observations [15]. In our model the unknown coefficients instead describe an independent waveform at every retained frequency, and the nonlinear parameters are source positions.

Candidate selection also has relevant precedents. Oh and Un use sequential MUSIC initialization before AP updates [11]. LAOLS retains discrete atom candidates and completes a tentative support before comparing its residual [4]. Here, a conditional spectrum holds the first position fixed, while continuous joint refinement allows both positions to move. The two operations can rank branches differently. Retaining several conditional candidates delays the decision until after continuous fitting, without repeating the full spatial scan for each branch.

The treatment of unknown waveforms affects refinement cost as well. Variable projection eliminates linear coefficients before optimizing nonlinear parameters [6]. We apply this principle to local block-NOMP position updates and examine the curvature term removed by waveform re-estimation. The paper makes two methodological contributions:

1. MCJ-AP compares conditional candidates after joint continuous refinement. Candidate retention improves initialization coverage, while joint updates reduce the cost of processing the retained branches at a common stopping accuracy.
2. VP-BNOMP retains residual discovery and cyclic block updates, but eliminates the current source waveform during each local position trial. A curvature analysis and a paired comparison with the same acceptance objective isolate its effect on repeated small updates.

Experiments compare eight method families on a common 240-observation subset, examine the principal methods on 2,400 observations, and test frozen configurations on 480 observations at new source positions. The receiver coordinates remain fixed. These experiments distinguish candidate recovery from local computational savings.

2. Observation model and localization objective

2.1. Synchronized wideband observations

Let $\boldsymbol{r}_l \in \mathbb{R}^2$ denote station l, $l = 1, \dots, L$, and let $\boldsymbol{p}_k \in \Omega \subset \mathbb{R}^2$ denote source k, $k = 1, \dots, K$. We assume static direct-path propagation, synchronized station timing and frequency, calibrated relative receiver responses, and matched antenna polarization with equal directional responses. The experiments use $L = 4$ and known $K = 2$. Define

$$d_{lk} = \|\boldsymbol{p}_k - \boldsymbol{r}_l\|_2, \qquad \tau_{lk} = d_{lk}/c_0, \qquad \beta_{lk} = d_{lk}^{-1}, \tag{1}$$

where c_0 is propagation speed. The amplitude follows free-space attenuation: for matched isotropic antennas, $P_\mathrm{r}/P_\mathrm{t} = (\lambda/4\pi d)^2$ [7]. At a fixed wavelength λ, the common factor $\lambda/(4\pi)$ is absorbed into the unknown waveform. The relative station amplitudes are therefore proportional to $1/d$.

The propagation phase is specified by an ideal baseband pure-delay model. Receiver calibration removes station responses; it does not establish this propagation assumption. Both generator and estimators use the same baseband model, without an RF carrier phase or random link gains. With sampling rate f_s and N samples,

$$x_l[n] = \sum_{k=1}^{K} \beta_{lk} s_k(n/f_s - \tau_{lk}) + w_l[n], \quad 0 \le n < N, \tag{2}$$

where s_k includes unknown transmitted power and the common amplitude factor. The source waveforms are independent. Noise samples are independent across stations and time, independent of the sources, and distributed as $\mathcal{CN}(0, \sigma_t^2)$.

The DFT is normalized by $1/N$. After retaining the common processing band $\mathcal{F}$, with $F = |\mathcal{F}|$ frequencies, the observation at frequency f is

$$\boldsymbol{x}_f = \boldsymbol{A}_f(\boldsymbol{P})\boldsymbol{s}_f + \boldsymbol{n}_f, \tag{3}$$

$$[\boldsymbol{a}_f(\boldsymbol{p})]_l = d_l(\boldsymbol{p})^{-1} \exp[-\mathrm{j}2\pi f d_l(\boldsymbol{p})/c_0], \tag{4}$$

$$\boldsymbol{A}_f(\boldsymbol{P}) = [\boldsymbol{a}_f(\boldsymbol{p}_1), \ldots, \boldsymbol{a}_f(\boldsymbol{p}_K)].$$

Here $\boldsymbol{x}_f, \boldsymbol{n}_f \in \mathbb{C}^L$, $\boldsymbol{s}_f \in \mathbb{C}^K$, and $\boldsymbol{P} = [\boldsymbol{p}_1, \ldots, \boldsymbol{p}_K] \in \mathbb{R}^{2\times K}$. Frequency f is the signed baseband frequency and $\mathrm{j}^2 = -1$. The common-window approximation neglects station-dependent edge differences [5]; a finite-window diagnostic is provided in Supplement S2. The retained noise variance remains $\sigma_f^2 = \sigma_t^2/N$. Band selection changes neither DFT normalization nor this variance.

Unit-norm responses $\boldsymbol{q}_f = \boldsymbol{a}_f/\|\boldsymbol{a}_f\|_2$ may be used computationally. A common scaling of one response column is absorbed by its free coefficient and preserves relative station amplitudes. The implementation protects distances below 1 m; all simulated sources lie far outside that region. Superscripts H, T and $\dagger$ denote conjugate transpose, transpose and pseudoinverse, respectively.

Let P_{lk} be the time-averaged power of isolated source k at station l, and let B be nominal signal bandwidth. We define

$$\mathrm{SNR}_{k,\mathrm{in}} = 10\log_{10}\frac{L^{-1}\sum_l P_{lk}}{\sigma_t^2 B/f_s}, \qquad \Delta = 10\log_{10}\frac{\sum_l P_{l1}}{\sum_l P_{l2}}. \tag{5}$$

Source 2 and the noise are fixed when scaling source 1. Thus changing Δ leaves the weak-source SNR unchanged. For $\Delta > 0$ the sources are called strong and weak; at $\Delta = 0$ their identities remain A and B. Local power differences at individual stations need not equal Δ.

2.2. Eliminating the unknown waveforms

Treat the frequency coefficients as deterministic unknowns. Under the independent equal-variance complex Gaussian noise model, maximum likelihood is equivalent to least squares. For observation energy $E_x = \sum_f \|\boldsymbol{x}_f\|^2 > 0$, define

$$\mathcal{L}(\boldsymbol{P}, \{\boldsymbol{s}_f\}) = E_x^{-1} \sum_f \|\boldsymbol{x}_f - \boldsymbol{A}_f(\boldsymbol{P})\boldsymbol{s}_f\|^2, \tag{6}$$

$$\widehat{\boldsymbol{s}}_f(\boldsymbol{P}) = \boldsymbol{A}_f(\boldsymbol{P})^\dagger \boldsymbol{x}_f, \tag{7}$$

$$J(\boldsymbol{P}) = E_x^{-1} \sum_f \|\boldsymbol{\Pi}^\perp_{\boldsymbol{A}_f} \boldsymbol{x}_f\|^2, \quad \boldsymbol{\Pi}^\perp_{\boldsymbol{A}_f} = \boldsymbol{I} - \boldsymbol{A}_f \boldsymbol{A}_f^\dagger. \tag{8}$$

Substitution of the least-squares waveforms yields the position-only objective J [6]. It measures the fraction of observation energy unexplained by the position set. Two-source localization then involves four real position variables, with coefficients fitted at every trial. The single-source score is

$$Q_0(\boldsymbol{p}) = E_x^{-1} \sum_f \frac{|\boldsymbol{a}_f(\boldsymbol{p})^H \boldsymbol{x}_f|^2}{\|\boldsymbol{a}_f(\boldsymbol{p})\|^2}. \tag{9}$$

For one source, $J = 1 - Q_0$. Taking two peaks from Q_0 does not jointly fit two sources; the subsequent algorithms use a multiple-source objective to account for their interaction.

3. Candidate retention and waveform-eliminated refinement

3.1. Conditional search and MCJ-AP

Given responses $\boldsymbol{B}_f$ at already selected positions, project both the data and trial response:

$$\boldsymbol{r}_f = \boldsymbol{\Pi}^\perp_{\boldsymbol{B}_f} \boldsymbol{x}_f, \qquad \boldsymbol{u}_f(\boldsymbol{p}) = \boldsymbol{\Pi}^\perp_{\boldsymbol{B}_f} \boldsymbol{a}_f(\boldsymbol{p}). \tag{10}$$

The conditional AP score is

$$Q_{\mathrm{AP}}(\boldsymbol{p} \mid \boldsymbol{B}) = E_x^{-1} \sum_f \frac{|\boldsymbol{u}_f(\boldsymbol{p})^H \boldsymbol{r}_f|^2}{\|\boldsymbol{u}_f(\boldsymbol{p})\|^2}. \tag{11}$$

For $\boldsymbol{u}_f \neq 0$, orthogonal projection gives

$$\boldsymbol{\Pi}^{\perp}_{[\boldsymbol{B}_f,\boldsymbol{a}_f]} = \boldsymbol{\Pi}^{\perp}_{\boldsymbol{B}_f} - \frac{\boldsymbol{u}_f \boldsymbol{u}_f^H}{\|\boldsymbol{u}_f\|^2}, \qquad J_{\mathrm{B}} - J([\boldsymbol{B}, \boldsymbol{a}(\boldsymbol{p})]) = Q_{\mathrm{AP}}, \tag{12}$$

where $J_{\mathrm{B}} = E_x^{-1} \sum_f \|\boldsymbol{r}_f\|^2$. This identity connects conditional search to joint least squares [5]: with existing positions fixed, the highest conditional score gives the greatest residual reduction. A zero projected response contributes zero. Numerical rank and denominator safeguards are specified in Supplement S3.

MCJ-AP first maximizes Q_0 on the common grid $\mathcal{G}$, then continuously refines the first source to $\boldsymbol{p}_1^{(0)}$. With that position fixed, it scans Q_{AP} and retains M local maxima $\boldsymbol{c}_m$ separated by at least Δ_c. Each supplies a joint initial position pair:

$$\widehat{\boldsymbol{P}}^{(m)} = \mathrm{LocalMin}_{\boldsymbol{P}\in\Omega^2} J(\boldsymbol{P}; [\boldsymbol{p}_1^{(0)}, \boldsymbol{c}_m]), \qquad m^* = \arg\min_m J(\widehat{\boldsymbol{P}}^{(m)}). \tag{13}$$

The output is $\widehat{\boldsymbol{P}}^{(m^*)}$. AP supplies conditional initialization; joint refinement subsequently adjusts both positions. Conditional grid ranking is therefore replaced by ranking after continuous fitting. A larger nested candidate set cannot increase the selected final residual when existing branch solutions remain unchanged; its position error must still be assessed empirically.

Joint refinement uses bounded L-BFGS-B [1]. Write $\boldsymbol{\theta} = \mathrm{vec}(\boldsymbol{P})$ and $\boldsymbol{e}_f = \boldsymbol{x}_f - \boldsymbol{A}_f \widehat{\boldsymbol{s}}_f$. When $\boldsymbol{A}_f$ is locally full column rank, $\boldsymbol{A}_f^H \boldsymbol{e}_f = 0$ eliminates the coefficient derivative from

$$\frac{\partial J}{\partial \theta_i} = -\frac{2}{E_x} \mathrm{Re} \sum_f \boldsymbol{e}_f^H \frac{\partial \boldsymbol{A}_f}{\partial \theta_i} \widehat{\boldsymbol{s}}_f. \tag{14}$$

For $\boldsymbol{v}_l = (\boldsymbol{p} - \boldsymbol{r}_l)/d_l$, the response derivative is

$$\nabla_{\boldsymbol{p}} a_l(f, \boldsymbol{p}) = a_l(f, \boldsymbol{p})(-d_l^{-1} - \mathrm{j}2\pi f / c_0)\boldsymbol{v}_l. \tag{15}$$

The two terms account for amplitude and delay. L-BFGS-B approximates joint curvature using position and gradient differences; it does not explicitly evaluate an exact cross-source Hessian.

Algorithm 1 MCJ-AP

Require: Observations, station positions, grid $\mathcal{G}$, candidate count M

1: Maximize Q_0 and refine the first position continuously.
2: Scan Q_{AP} conditional on this position.
3: Retain M separated local maxima in descending score order.
4: **for** each retained candidate **do**
5: Jointly minimize J from the resulting position pair, fitting all coefficients at every trial.
6: **end for**
7: Return the branch with the smallest final J.

3.2. Waveform-eliminated block NOMP

VP-BNOMP preserves residual discovery and cyclic position updates [10, 14]. During the update of source j, hold the fitted contributions of the other sources fixed:

$$\boldsymbol{y}_{j,f} = \boldsymbol{x}_f - \sum_{k \neq j} \boldsymbol{a}_f(\boldsymbol{p}_k)\widehat{s}_{k,f}. \tag{16}$$

For a trial position $\boldsymbol{p}$, eliminate the current source coefficients:

$$\widehat{q}_f(\boldsymbol{p}) = \frac{\boldsymbol{a}_f(\boldsymbol{p})^H \boldsymbol{y}_{j,f}}{\|\boldsymbol{a}_f(\boldsymbol{p})\|^2}, \tag{17}$$

$$\phi_j(\boldsymbol{p}) = E_x^{-1} \sum_f \left(\|\boldsymbol{y}_{j,f}\|^2 - \frac{|\boldsymbol{a}_f(\boldsymbol{p})^H \boldsymbol{y}_{j,f}|^2}{\|\boldsymbol{a}_f(\boldsymbol{p})\|^2} \right). \tag{18}$$

Here q_f is a complex scalar, distinct from the response vector $\boldsymbol{q}_f$. A Newton direction is computed from the gradient and Hessian of ϕ_j. Backtracking re-estimates q_f at each trial and accepts a nonincreasing local residual. The accepted contribution and explicit residual are updated together. Common least squares then re-estimates all coefficients after cyclic refinement following each source addition.

New positions are discovered by scanning the current joint residual $\boldsymbol{r}_f$ with

$$Q_{\mathrm{R}}(\boldsymbol{p}) = E_x^{-1} \sum_f \frac{|\boldsymbol{a}_f(\boldsymbol{p})^H \boldsymbol{r}_f|^2}{\|\boldsymbol{a}_f(\boldsymbol{p})\|^2}. \tag{19}$$

After common least squares, $\boldsymbol{r}_f$ is orthogonal to existing responses. The AP and residual scores then share a numerator, but use projected and unprojected response norms, respectively. Their

discovery mechanisms can select different regions even though both eventually perform least-squares fitting.

Algorithm 2 VP-BNOMP

Require: Observations, grid, source count K, refinement parameters

Initialize an empty source set and the observation residual.
for $k = 1, \dots, K$ **do**
    Maximize Q_{R}, locally refine with ϕ_j, and add the source.
    repeat
        for each discovered source j **do**
            Fix the other source contributions and refine ϕ_j with safeguarded Newton updates and waveform re-estimation.
        end for
    until the cyclic energy condition or safety limit is met
    Jointly re-estimate all source coefficients and update the residual.
end for
Return the discovered positions.

3.3. Curvature and computational cost

For one local update, collect the two position coordinates in $\boldsymbol{z}$ and the real and imaginary coefficient parts in $\boldsymbol{u} \in \mathbb{R}^{2F}$. Let $\mathcal{L}_{\mathrm{loc}}(\boldsymbol{z}, \boldsymbol{u})$ denote its normalized least-squares loss and $\widehat{\boldsymbol{u}}(\boldsymbol{z})$ its coefficient minimizer. If the response is twice continuously differentiable and this minimizer is locally unique, differentiating $\nabla_u \mathcal{L}_{\mathrm{loc}} = 0$ yields

$$\frac{\partial \widehat{\boldsymbol{u}}}{\partial \boldsymbol{z}} = -H_{uu}^{-1} H_{uz}, \qquad H_{\mathrm{VP}} = H_{zz} - H_{zu} H_{uu}^{-1} H_{uz}. \tag{20}$$

All Hessian blocks are evaluated at $(\boldsymbol{z}, \widehat{\boldsymbol{u}}(\boldsymbol{z}))$. The position, coefficient and mixed blocks have dimensions 2×2, $2F \times 2F$ and $2 \times 2F$, respectively. The reduced gradient equals the position gradient at the coefficient optimum [6]. Since the coefficient subproblem has positive-definite H_{uu},

$$H_{zz} - H_{\mathrm{VP}} = H_{zu} H_{uu}^{-1} H_{uz} \succeq 0. \tag{21}$$

Part of a response change caused by moving a source can be compensated by changing its waveform. Fixed-coefficient curvature retains this component; reduced curvature removes it. In positive-curvature directions with strong coupling, the former can produce small position steps.

At an interior local minimum, suppose H_{zz} and H_{VP} are positive definite and full Newton steps are used. For position error $\boldsymbol{\epsilon}$ relative to that local minimum, a fixed-curvature update of the reduced gradient has linearization

$$\boldsymbol{\epsilon}^{+} \approx (\boldsymbol{I}_2 - \boldsymbol{H}_{zz}^{-1} \boldsymbol{H}_{\mathrm{VP}})\boldsymbol{\epsilon}. \tag{22}$$

By (21), the iteration eigenvalues lie in $[0, 1)$ and approach one when reduced curvature is small relative to fixed-coefficient curvature. Exact reduced Newton curvature removes this linear term. Backtracking, safeguards and the outer source updates affect actual cost, which is measured separately from this local analysis.

The coefficient Hessian is block diagonal across frequencies, with 2×2 blocks. The position Hessian can thus be accumulated frequency by frequency without forming a dense $2F \times 2F$ matrix. Both local curvature calculations scale linearly with F; the efficiency gain concerns actual steps and cycles rather than a lower order in F. A joint coefficient fit costs approximately $\mathcal{O}(F(LK^2 + K^3))$. MCJ-AP requires two grid scans and M local solves, approximately $\mathcal{O}(FGLK + T_M F(LK^2 + K^3))$, where G is grid size and T_M counts all branch evaluations. VP-BNOMP adds cyclic local updates to residual scans; their cost grows with the observed cycle and derivative counts. Full method-wise complexity is provided in Supplement S4.

For fixed source count, accepted MCJ-AP steps and VP-BNOMP block updates do not increase their respective fitting residuals; common coefficient re-estimation also cannot increase the residual. The resulting nonnegative objective sequences converge. This property concerns objective values, not global position convergence. AP checks a projected joint gradient, whereas VP-BNOMP checks cyclic energy change. The mechanism experiments retain the corresponding matched stopping rules.

4. Simulation results

4.1. Design, baselines and evaluation

The stations are located at $(500, 0)$, $(0, 500)$, $(-500, 0)$ and $(0, -500)$ m. Their maximum baseline is $R = 1,000$ m and the search region is $[-1,000, 1,000]^2$ m^2. Table 1 lists the common acquisition and search settings. Source positions are continuous. The two independent BPSK

waveforms have independent symbol sequences, initial phases and timing, and the same center frequency and bandwidth.

Table 1

Common acquisition and search settings.

Quantity	Setting
Stations and sources	Four synchronized single-antenna stations; known $K = 2$
Propagation	Relative amplitude $1/d$ and baseband delay; $c_0 = 299792458$ m/s
Signal	BPSK; RRC shaping, 10 MHz symbols, roll-off 0.25; 12.5 MHz bandwidth
Acquisition	50 MHz, 8,192 samples, 163.84 μs
Processing band	$[-7, 7]$ MHz; 2,293 full-record DFT bins
Power	Weak-source mean in-band SNR 6 dB; $\Delta = 0, 3, 6, 9, 12, 15$ dB
Search	Common 20 m grid, 101×101 points; continuous refinement
Candidates	3×3 local maxima, minimum separation 30 m; $M = 1$–5

Within each source configuration and realization, all six power gaps reuse the same waveforms and noise. Only source A is scaled. The band is fixed in advance, includes 0.75 MHz guards around the nominal signal band, and is shared by every method; no per-source spectrum or per-station RMS normalization is supplied.

Twenty source configurations cover both sources inside the station convex hull, A outside/B inside, A inside/B outside, and both outside, with five configurations per category (Fig. 1). Generation excludes sources within 150 m of a station, 100 m of the center or 50 m of a station-pair line, and requires at least 300 m between sources. Positions are not selected by localization performance. All methods are evaluated on two preassigned realizations per configuration and all power gaps, giving 240 observations. Core methods use 20 realizations per configuration, giving 2,400 observations that include the common 240. Frozen configurations are additionally evaluated on 16 new source configurations, five realizations each and six gaps, giving 480 observations. The receiver geometry is unchanged. Coordinates and seeds are provided in Supplement S1.

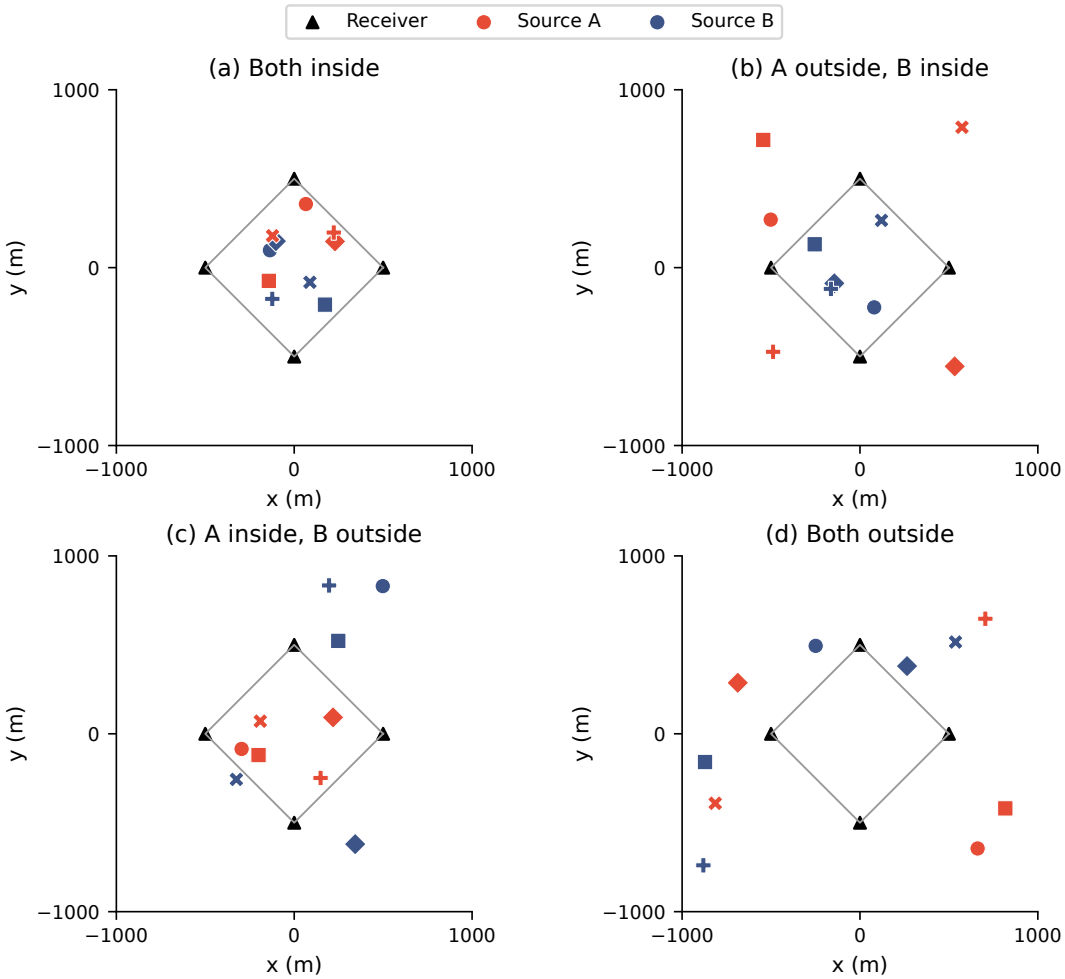


Figure 1: Twenty source configurations, five per category. Black triangles denote stations. Matching shapes pair source A (red) and B (blue) within each panel; the diamond is the station convex hull.

All methods share IQ data, propagation information, numerical precision and the initial grid. Baselines include single-source LS peaks, wideband MUSIC, MVDR-known, full AP, fixed-coefficient BNOMP, shared-power SBL, multistart joint ML and GA-based joint ML. MUSIC uses four segments and noise-subtracted observed-power weights; MVDR-known uses eight segments and relative loading 0.01. MVDR-known is distinguished from the original DPD-HR formula. Full AP repeats conditional scans and continuous refinement. Multistart ML uses four unique spectral starts and four Sobol starts; GA uses 128 individuals and at most 200 generations. SBL uses at most 300 iterations and its registered off-grid update. Their adapted objectives, parameters and stopping conditions are given in Supplement S2–S3.

AP branch refinements share projected-gradient tolerance 10^{-8} on the normalized objective, no objective-difference stopping term, and a 100-iteration local limit. VP-BNOMP and its fixed-coefficient reference use cyclic energy tolerance 10^{-8}, at most five Newton steps per local call, a 50 m step cap, a 0.01 m local small-step exit and a 10,000-cycle safety limit. The different gradient and energy conditions are not equivalent optimization accuracies. Complete-method comparisons use their frozen native rules; joint-refinement and curvature comparisons match the relevant stopping conditions explicitly.

For every observation, one squared-distance assignment [9] determines errors for all reported position metrics:

$$\pi^* = \arg\min_{\pi} \sum_{k} \|\widehat{\boldsymbol{p}}_{\pi(k)} - \boldsymbol{p}_k\|^2, \quad e_{ik} = \|\widehat{\boldsymbol{p}}_{i,\pi^*(k)} - \boldsymbol{p}_{ik}\|. \tag{23}$$

RMSE is $\sqrt{n^{-1}\sum_i e_{ik}^2}$ and the empirical CDF is $n^{-1}\sum_i \mathbf{1}(e_{ik} \le t)$. Both include all observations. Two-source success requires $\max_k e_{ik} \le 20$ m, a prescribed tolerance of $0.02\boldsymbol{R}$ distinct from grid spacing. Invalid outputs remain failures in the denominator; all reported batches returned two finite positions.

Complete runtime includes the DFT, band selection, transfer, search, coefficient estimation, refinement and necessary output checks, with GPU synchronization. Adapter trajectory recording and final diagnostics are included where present; input-file reading, disk writes and offline statistics are excluded. Warm-up is separate. Runtime comparisons use the same batch for speedup ratios. The environment is Windows, Python 3.10.20, NumPy 2.2.6, PyTorch 2.8.0+cu129, an RTX 4070 Ti SUPER and 20 CPU threads, with double-precision responses and objectives. Resampling retains the six gaps of one realization as a group. Candidate-cost and curvature intervals condition on fixed layouts; broader hierarchical intervals are identified separately below.

4.2. Common-data comparison and expanded evaluation

Table 2 compares all methods on the same 240 observations. MCJ-AP matches the complete recovery of full AP and SBL at substantially lower complete cost. SBL has a wider weak-source

error distribution. The two BNOMP versions recover the same 228 observations, with lower mean cost for the waveform-eliminated version.

Table 2

Common 240 observations. RMSE in m; runtime mean, median and P95 in s. Every metric includes all observations.

Method	Success/240	RMSE A	RMSE B	Mean	Median	P95
LS peaks	38	191.982	845.184	0.2368	0.2181	0.3297
WB-MUSIC	151	93.515	646.892	0.2015	0.1893	0.2719
MVDR-known	75	320.684	804.082	0.1925	0.1807	0.2625
Full AP	240	0.876	1.596	8.9076	9.1483	12.2348
Fixed BNOMP	228	0.906	89.706	2.9416	1.4625	8.1989
VP-BNOMP	228	0.875	86.177	0.5907	0.3921	1.3971
Multistart ML	45	256.061	776.935	1.2227	1.0868	2.5595
GA-ML	185	279.056	501.231	8.0702	8.1102	9.0229
WB-SBL	240	1.096	3.743	44.1296	36.8237	97.7075
MCJ-AP ($M = 5$)	240	0.876	1.596	0.7081	0.6347	1.0232

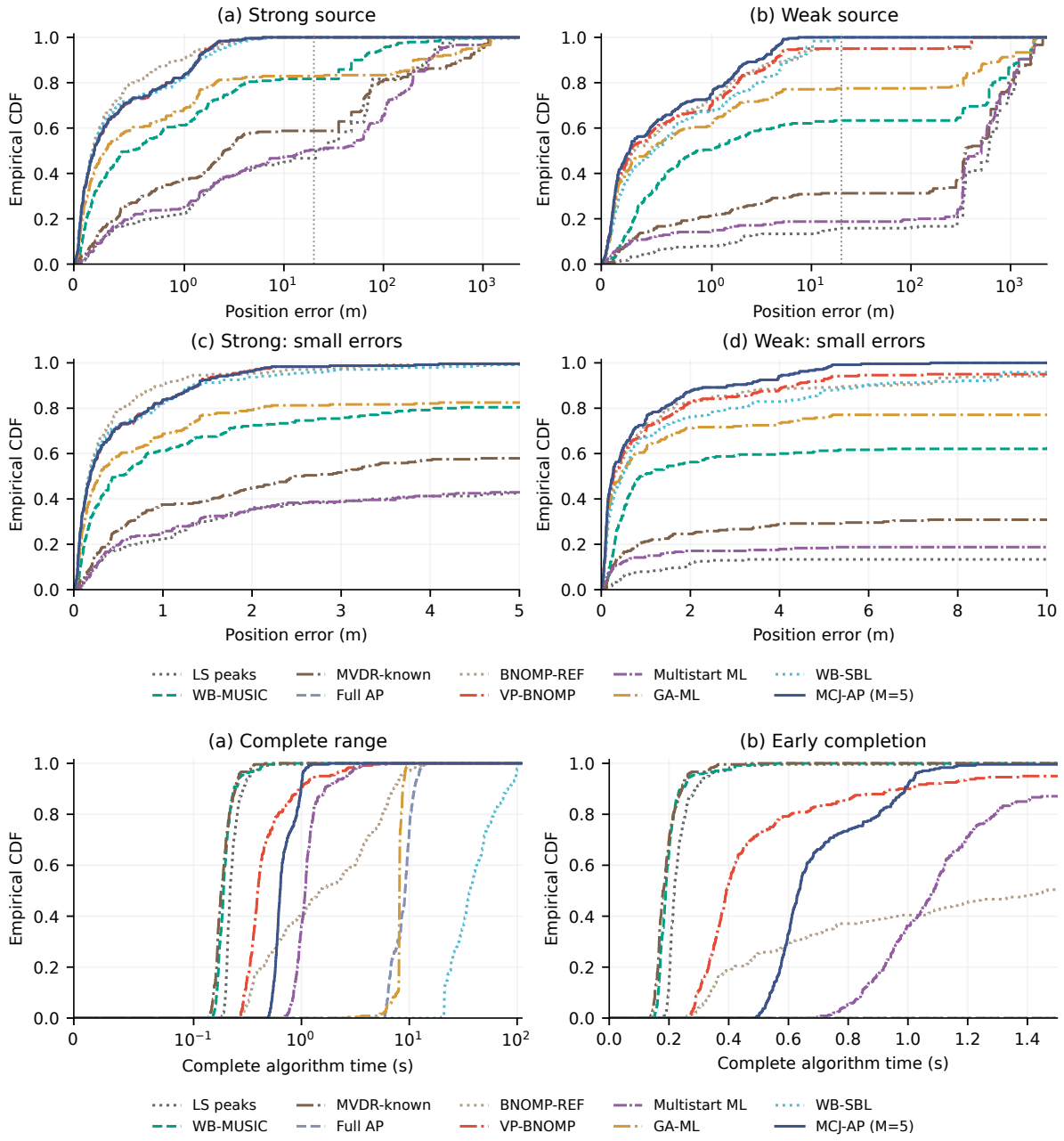


Figure 2: Error and complete-runtime CDFs on the common 240 observations. The error panels show full ranges and small-error enlargements; the time panels show the full range and early completion. Full-range axes are linear near zero and logarithmic beyond their thresholds. Vertical error lines indicate 20 m. All curves retain the full denominator.

Figure 2 distinguishes typical precision from large-error tails. The AP error curves nearly coincide. VP-BNOMP is accurate on most observations, but retains a distant weak-source tail

and occasional long runtimes. Spectral methods are fast, yet their failures involve large errors rather than small threshold crossings.

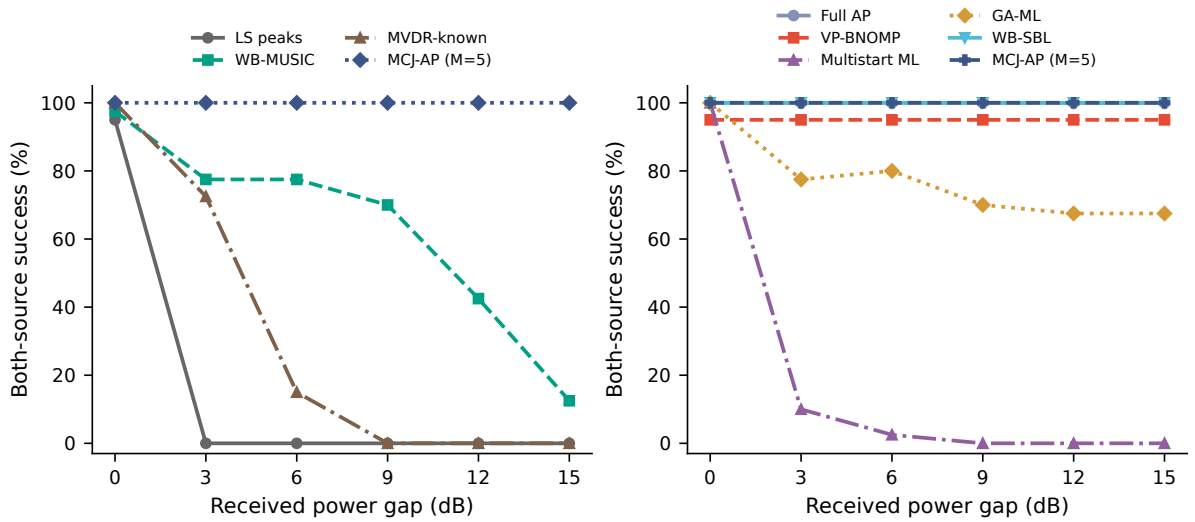


Figure 3: Power-gap sweep on the common subset, 40 observations per gap. Both panels use the same five-candidate AP reference.

As the power gap increases, the registered MUSIC and MVDR-known implementations lose recovery (Fig. 3). A window diagnostic on eight development observations reduced or removed the clean-data model residual without recovering the original failed method–observation pairs (Supplement S2). Finite-window mismatch alone therefore does not explain those tested failures. Full AP and multistart ML optimize the same joint objective but have very different outcomes, demonstrating the importance of initialization and search. GA reaches its generation limit in 208 observations; SBL reaches 300 iterations in 14. These outputs are included in the comparison.

Table 3

Expanded 2,400-observation evaluation. RMSE in m and complete time in s. VP-BNOMP timing is from a separate same-input batch; controlled speedups are evaluated in paired batches.

Method	Success/2400	RMSE A	RMSE B	Mean	Median	P95
AP ($M = 1$)	2280	0.897	107.265	0.3958	0.3700	0.5104
AP ($M = 2$)	2280	0.897	103.698	0.4704	0.4329	0.6340
MCJ-AP ($M = 5$)	2376	0.896	46.984	0.7180	0.6495	1.0267
VP-BNOMP	2278	0.892	74.193	0.5855	0.3693	1.2470

On 2,400 observations, five candidates increase AP success from 95% to 99% (Table 3). Hierarchical resampling gives a 95% interval of [96.83%,100%] for its success rate. The four-percentage-point increase over two candidates has interval [0,12.17] percentage points, reflecting concentration in one configuration. The common 240 subset and expanded cohort are reported separately rather than counted as independent observations.

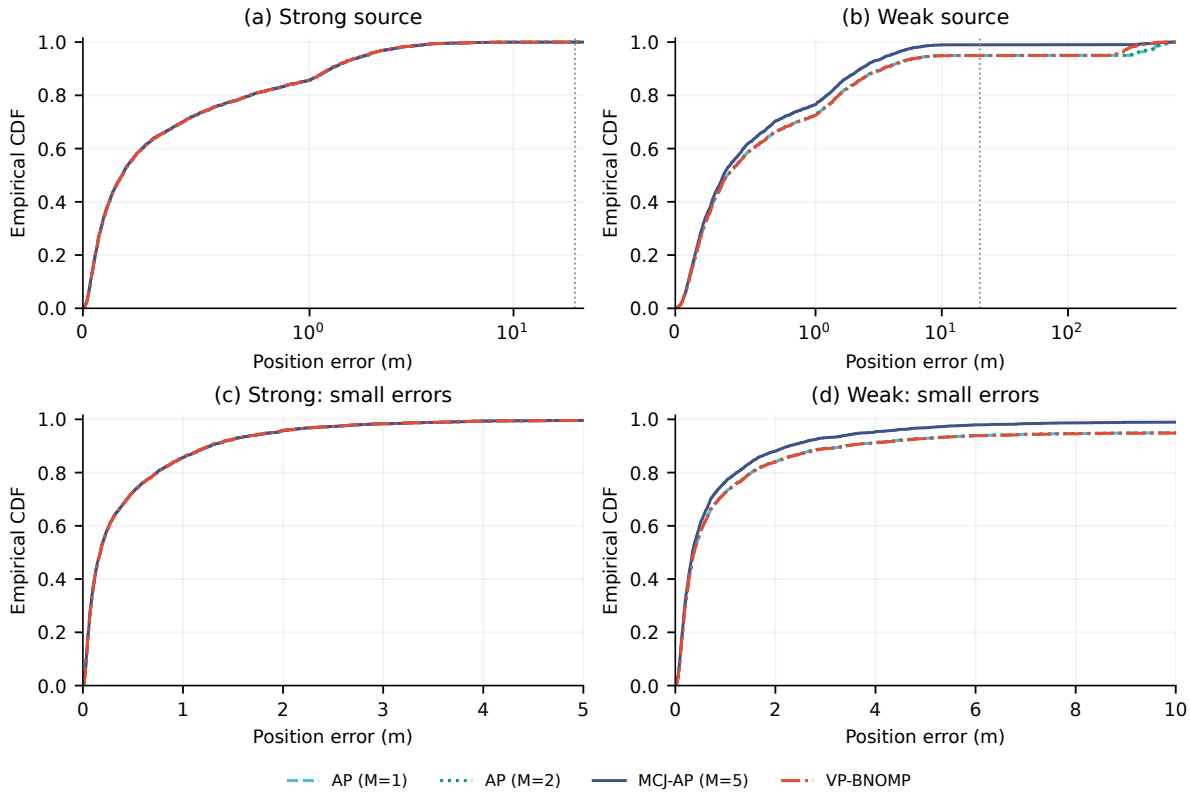


Figure 4: Strong- and weak-source error CDFs for all 2,400 core observations. Upper panels cover full ranges and lower panels enlarge small errors. All observations, including failures, remain in each curve.

The strong-source CDFs nearly coincide (Fig. 4). Small weak-source errors are also similar, whereas tails differ. Five-candidate AP has weak-source P95 of 3.85 m but RMSE of 46.98 m, showing the effect of a small number of large errors. Configuration 4 accounts for all 24 of its failures and over 99.7% of weak-source squared error in each power bin. Separate configuration and power-gap plots are given in Supplement S5; all configurations remain in the overall statistics.

4.3. Candidate coverage and joint-refinement cost

The 96 AP recoveries occur in configuration 4 and involve 18 of its 20 waveform/noise realizations. Single-candidate AP fails on all 120 observations there, whereas five candidates recover 96. No observation is changed from success to failure in the other configurations. Saved branch results show that exactly 2,376 observations have at least one successful refined branch,

and final residual selection chooses a successful branch in every such observation. The remaining 24 have no successful branch among the five refined candidates.

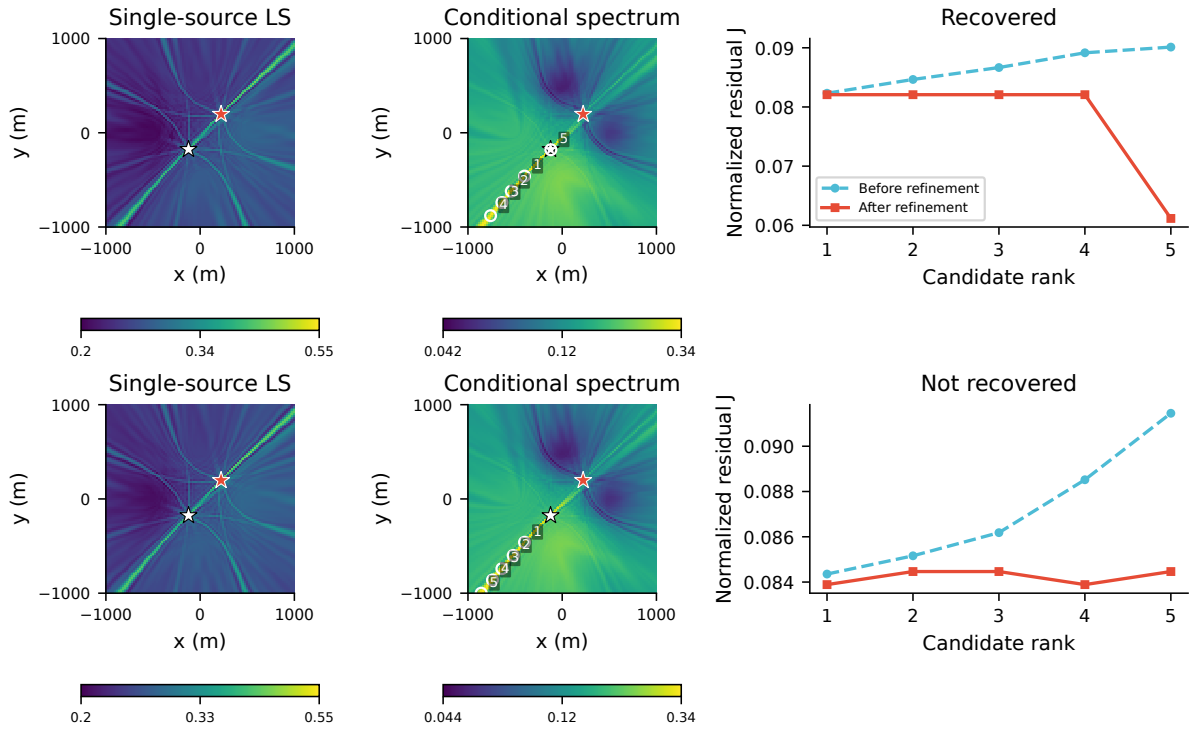


Figure 5: Two post hoc cases from configuration 4 at 0 dB. Columns show the single-source spectrum, conditional spectrum and pre-/post-refinement branch residuals. The upper case is recovered by the fifth candidate; the lower case has no effective candidate. Lines in the right panels connect branch indices, not optimization trajectories.

In the upper case of Fig. 5, the first four branches enter a distant region; the fifth gives a lower final residual and approximately 0.09 m weak-source error. In the lower case, even the nearest candidate is about 397 m from source B and refinement cannot recover it. These cases explain why retaining candidates can help while local refinement alone cannot repair missing coverage.

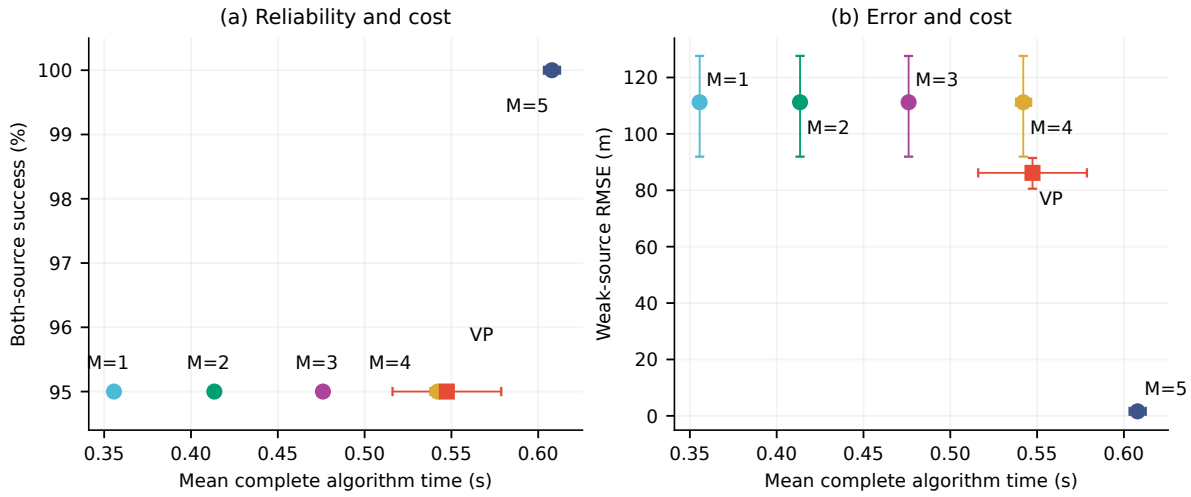


Figure 6: Candidate count versus complete mean runtime, success and weak-source RMSE on 240 observations, measured in one interleaved batch. Error bars are 95% paired intervals resampling realizations within fixed layouts while retaining the six gaps together.

In this batch, $M = 1$–4 each succeeds on 228 observations; mean runtime increases from 0.3556 to 0.5420 s and weak-source RMSE remains about 111.2 m. VP-BNOMP has 228 successes, 86.18 m weak-source RMSE and 0.5474 s mean runtime. Five-candidate AP succeeds throughout at 0.6079 s and 1.60 m weak-source RMSE. Thus VP reduces some severe errors relative to low-candidate AP, while five candidates improve recovery. Saved full-cohort prefixes give 2,281 and 2,289 successes for three and four candidates; those are position-only replays, not full-cohort runtime measurements.

Table 4

Paired alternating and joint refinement of the same five candidates on 2,400 observations. Both selected outputs meet the same projected-gradient tolerance.

Refinement	Success	RMSE B/m	Mean time/s	J calls	Gradient calls
Alternating	2376	46.983	1.0840	204.21	198.21
Joint	2376	46.984	0.7441	120.36	114.36

Table 4 isolates the cost of refinement from candidate coverage. Success/failure identities agree observation by observation, strong-source RMSE is about 0.896 m, and weak-source

P95 is about 3.85 m for both. Joint refinement reduces mean complete time by 31.36% and gradient calls by 42.30%. The paired hierarchical 95% interval for joint-minus-alternating time is $[-0.3691, -0.3121]$ s. Every configuration has lower mean cost, with savings from 11.42% to 48.04%. Two unselected alternating branches reach the local safety limit and remain included. Candidate retention supplies additional starts; joint refinement processes them with fewer calls at matched accuracy.

4.4. Isolating waveform-eliminated curvature

The common-data fixed-coefficient/VP comparison in Table 2 changes both local curvature and the trial acceptance objective, reducing mean runtime by 79.92%. To isolate curvature, a separate paired batch applies the same waveform-eliminated objective ϕ_j to trial acceptance. FC–VP computes directions with H_{zz} and VP–VP with H_{VP}. Both use the same current conditional coefficient estimate, gradient, discovery sequence, safeguards, backtracking and cyclic energy stopping rule.

Table 5

Curvature comparison with a common waveform-eliminated acceptance objective, 240 paired observations. Update and cycle counts are means per observation.

Version	Success	RMSE A/m	RMSE B/m	Mean time/s	Updates	Cycles
FC–VP	228/240	0.906	89.706	5.574	1498.82	810.90
VP–VP	228/240	0.875	86.177	0.905	135.55	47.97

Success/failure identities agree and all outputs stop under the native energy rule. Mean complete runtime falls by 83.77%, with fixed-layout paired 95% interval [81.73%,85.58%]. Median time falls from 2.836 to 0.567 s and P95 from 16.161 to 2.538 s. VP–VP is faster in 170 of 240 observations: the mean saving is driven especially by reductions in slow cases. The associated update and cycle reductions support the curvature mechanism in (22).

Recording and final checks average 0.0254 and 0.0059 s in the two versions, much smaller than the cost difference. Historical and current VP paths agree in positions, objectives, configurations, counts and saved trajectories, despite different absolute wall times. Speedup percentages are therefore computed within their respective paired batches. The 79.92% result concerns

the complete local-update change; 83.77% concerns curvature under common acceptance. A common outer energy rule does not impose equal final joint-gradient accuracy. The interpretation is an observed update-cost benefit, with localization quality reported alongside it.

4.5. Frozen methods at new source positions

Table 6 and Fig. 7 assess the 16 new source configurations. Both AP variants recover all 480 observations with nearly identical errors, while five candidates cost approximately 1.96 times as much as one. The first branch is already effective, so additional branches do not increase success. Together with configuration 4, this identifies candidate retention as a cost-for-coverage choice whose recovery benefit depends on initialization.

Table 6

New source configurations, 480 observations; receiver positions unchanged. RMSE in m and complete runtime in s.

Method	Success/480	RMSE A	RMSE B	Mean	Median	P95
AP ($M = 1$)	480	0.958	2.194	0.461	0.456	0.545
MCJ-AP ($M = 5$)	480	0.958	2.194	0.906	0.893	1.078
VP-BNOMP	390	202.882	206.261	1.051	0.629	4.245

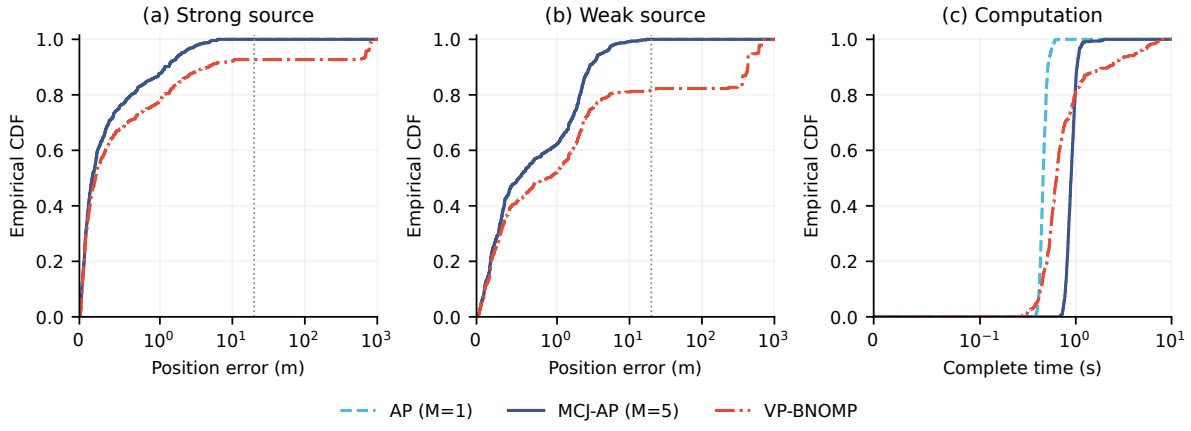


Figure 7: Error and complete-runtime CDFs on the 480 new observations. AP error curves nearly overlap. All observations are retained; error reference lines mark 20 m. Full-range axes are linear near zero and logarithmic beyond the stated thresholds.

VP-BNOMP has 90 failures concentrated in new configurations 8, 9, 14 and 15, with 30, 25, 30 and 5 failures. Each gap has 15 failures; 55 involve only the weak source, 33 both sources and two only the strong source. Every failed observation is already unsuccessful at the first saved full two-source state and remains so at later saved states. Its final fitting objective exceeds AP's, and none is truncated by the safety limit. These results establish configuration-dependent failures of the complete sequential procedure, despite its improved local update. The stored trajectory begins after the first two-source cycle and does not resolve whether discovery or a particular local update first lost the useful solution.

5. Conclusions

For unknown overlapping wideband sources, candidate coverage and position–waveform coupling affect different parts of direct localization. MCJ-AP retains conditional candidates, jointly refines each pair and selects the final residual. It raises two-source success from 95% to 99% on 2,400 observations, while matched-accuracy joint refinement reduces the cost of processing five candidates by 31.36%. VP-BNOMP removes waveform coupling from local curvature; a common-acceptance paired comparison reduces mean runtime by 83.77%, together with fewer local updates and cycles. On new source configurations, AP remains reliable but extra candidates provide no additional recovery, and VP retains configuration-dependent failures. Candidate retention therefore supplies conditional recovery benefits, whereas reduced curvature supplies a local efficiency benefit. The conclusions concern known two-source count, synchronized stations and the matched baseband propagation model. Candidate allocation and unknown source count provide directions for further work.

References

[1] Byrd, R.H., Lu, P., Nocedal, J., Zhu, C., 1995. A Limited Memory Algorithm for Bound Constrained Optimization. SIAM Journal on Scientific Computing 16, 1190–1208. doi:10.1137/0916069.

[2] Capon, J., 1969. High-resolution frequency-wavenumber spectrum analysis. Proceedings of the IEEE 57, 1408–1418. doi:10.1109/proc.1969.7278.

[3] Chan, Y., Ho, K., 1994. A simple and efficient estimator for hyperbolic location. IEEE Transactions on Signal Processing 42, 1905–1915. doi:10.1109/78.301830.

[4] Chatterjee, S., Sundman, D., Vehkapera, M., Skoglund, M., 2012. Projection-Based and Look-Ahead Strategies for Atom Selection. IEEE Transactions on Signal Processing 60, 634–647. doi:10.1109/TSP.2011.2173682.

[5] Chen, J.C., Hudson, R.E., Yao, K., 2002. Maximum-likelihood source localization and unknown sensor location estimation for wideband signals in the near-field. IEEE Transactions on Signal Processing 50, 1843–1854. doi:10.1109/tsp.2002.800420.

[6] Golub, G.H., Pereyra, V., 1973. The Differentiation of Pseudo-Inverses and Nonlinear Least Squares Problems Whose Variables Separate. SIAM Journal on Numerical Analysis 10, 413–432. doi:10.1137/0710036.

[7] International Telecommunication Union, 2024. Calculation of free-space attenuation. Recommendation ITU-R P.525-5. ITU Radiocommunication Sector. URL: https://www.itu.int/rec/R-REC-P.525-5-202411-I/en.

[8] Knapp, C., Carter, G., 1976. The generalized correlation method for estimation of time delay. IEEE Transactions on Acoustics, Speech, and Signal Processing 24, 320–327. doi:10.1109/tassp.1976.1162830.

[9] Kuhn, H.W., 1955. The Hungarian method for the assignment problem. Naval Research Logistics Quarterly 2, 83–97. doi:10.1002/nav.3800020109.

[10] Mamandipoor, B., Ramasamy, D., Madhow, U., 2016. Newtonized Orthogonal Matching Pursuit: Frequency Estimation Over the Continuum. IEEE Transactions on Signal Processing 64, 5066–5081. doi:10.1109/tsp.2016.2580523.

[11] Oh, S.K., Un, C.K., 1991. Fast initialization procedure of the alternating projection algorithm for maximum likelihood localization of multiple sources. Signal Processing 25, 381–389. doi:10.1016/0165-1684(91)90122-Y.

[12] Tirer, T., Weiss, A.J., 2016. High Resolution Direct Position Determination of Radio Frequency Sources. IEEE Signal Processing Letters 23, 192–196. doi:10.1109/lsp.2015.2503921.

[13] Weiss, A.J., Amar, A., 2005. Direct Position Determination of Multiple Radio Signals. EURASIP Journal on Advances in Signal Processing 2005, 653549. doi:10.1155/asp.2005.37.

[14] Yang, Y., Yang, Y., Chu, Z., Shen, L., 2022. Multi-frequency synchronous two-dimensional off-grid compressive beamforming. Journal of Sound and Vibration 517, 116549. doi:10.1016/j.jsv.2021.116549.

[15] Zeng, Y., Han, M., Li, X., Li, T., 2025. QNOMP: A Quasi-Newton-Based Joint Multipath Optimization Method for Super-Resolution Channel Estimation and Extrapolation. IEEE Access 13, 212652–212671. doi:10.1109/access.2025.3645128.